Back **Week Five Quiz**
Graded Assignment

Due Mar 15, 11:59 PM CET

1. In the jury version of hypothesis testing, the alternative hypothesis is: 1 point

- H_0 :
not guilty.
- H_1 :
guilty

2. In hypothesis testing, a Type II error is: 1 point

- rejecting
H_0 when H_0 is true
- not
rejecting H_0 when H_0 is false

3. In hypothesis testing, the conditional probability
$P(H_0 \text{ not rejected} \mid H_0 \text{ is true}) =$ 1 point

- α
- β
- $1 - \alpha$
- $1 - \beta$

4. In general, we control for the probability of a Type I error
by choosing: 1 point

- $\alpha \in [0,\ 1]$ and so test at the $100\alpha\%$ significance level
- $\alpha \in [0,\ 1]$ and so test at the $100(1-\alpha)\%$ significance level

5. In hypothesis testing, the p-value decision rule is: 1 point

- reject
H_0 if the p-value is $\leq \alpha$
- reject
H_0 if the p-value is $\leq \alpha$
- reject
H_0 if the p-value is $\geq \alpha$
- reject H_1 if the p-value is $\geq \alpha$

6. Other things equal, as the effect size gets larger, the p-value: 1 point

- gets smaller
- stays the same
- gets larger

7. If a hypothesis test rejects H_0, then which of the following might have occurred? 1 point

- A Type I error
- A Type II error

8. Under the central limit theorem, when sampling from non-normal distributions the distribution of $\bar{X}$ tends to: 1 point

- a normal distribution with mean μ and variance σ^2, as $n \to \infty$
- a normal distribution with mean μ and variance σ^2/n, as $n \to \infty$

9. The sampling distribution of the sample proportion, $\hat{P}$, is: 1 point

- $N(\pi,\ \pi(1-\pi)/n)$ for any n
- $N(\pi,\ \pi(1-\pi)/n)$ as $n \to \infty$

10. A confidence interval for a proportion is given by: 1 point

- $\hat{p} \pm z \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
- $\hat{p} \pm z \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$